FINAL ASSESSMENT PROJECT

Module Details

Module Code	UFCFK1-15-0
Module Title	Fundamentals of Data Science
Module Tutors	Saurav Gautam
Year	2025/2026
Component/Element Number	Portfolio
Weighting	40%

Dates

Submission Date	**17-April-2026**
Submission Place	Backboard
Submission Time	23:59
Submission Notes	Submit Gitlab URL

Task 1 *(30: basic requirements + 10: bonus points for creativity)*

Objective:
Build a **Student Profile Management System** to handle student personal information, examination data, and extracurricular activities (ECA).

Your program must use four text files:

- **users.txt**: Stores registered users' personal information, including their roles (admin/student).
- **grades.txt**: Contains marks for five subjects for students.
- **eca.txt**: Details of extracurricular activities for each student.
- **passwords.txt**: Stores usernames and their passwords (use simple storage for learning purposes)

Basic Requirements:

1. Implement a login system that validates usernames and passwords.
    - Display an error and re-prompt for incorrect credentials.
    - Differentiate between admin and student roles upon login.
2. **Admin Role**:
    - Add new users with unique IDs.
    - Update, modify, or delete student records (personal, grades, and ECA).
    - Generate useful insights from the data (e.g., average grades per subject, most active students in ECA).

3. **Student Role**:
    - Update their profile information.
    - View personal details, ECA participation, and exam grades.
4. Ensure the program demonstrates:
    - Object-Oriented Programming (OOP) concepts (classes, inheritance, etc.).
    - Functions for modular design.
    - File handling for reading/writing data.
    - Exception handling for error-prone sections.
5. Creativity is rewarded for features like:
    - Visualizing data (e.g., plotting grades).
    - Improved user interface or additional functionalities.

Task 2 *(Optional Advanced Feature – 10 Bonus Points)*

Implement a Performance Analytics Dashboard:
For admin users, create an analytics module that provides:

- **Grade Trends**: Display trends for students' grades using charts.
- **ECA Impact**: Correlate ECA involvement with academic performance.
- **Performance Alerts**: Identify students performing below a specific threshold and suggest interventions.

Use libraries such as **matplotlib** or **pandas** for data visualization and analysis.

Submission Requirements:

- Submit the source code.
- Include sample input/output to demonstrate the functionality.
- Submit a written report describing the project, relevant flowcharts, test output screen screenshots.
- Work contribution and reflection of each member.
- Presentation slides.